



IT001-4-3 Internship (part 2)

Student Name : Adam Mohamed Bannani
 Student ID No : Tp073524
 Intake : January 2026
 Company Name : Caliber
 Academic Supervisor : Jack lai Yong Geat
 Start Date : 15/1/2026
 End Date : 15/5/2026

For submission, please fill up below:

Part 1	√ for yes and X for no	Part 2	√ for yes and X for no
Email sent to Academic Supervisor by the end of the first week of internship	yes	Industrial Placement Report	yes
APU Internship Supervisory Log Sheet signed by Academic Supervisor	yes	Weekly logbooks – signed and stamped by the company	yes
Weekly logbooks – signed and stamped by the company	yes	Report Clearance Form – signed and stamped by the company	yes
		Attendance record – signed and stamped by the company	yes
		Industrial Supervisor Assessment Form – signed and stamped by the company	yes
		Internship/Industrial Placement Visit Report Form – completed by Academic Supervisor	yes

Internship/Industrial Placement Report Clearance Form



Instruction to Student:

This form must be provided to the Industrial Supervisor for report clearance. Please ensure that the completed form is attached to your report for submission.

Student Details

Student Name	Adam Mohamed Bannani
Department	Industrial Design
Student ID	TP073524
Intake	March 2024
Student Passport/IC No	A28421283

Company Details

Company Name	Caliber
Contact Person (Industrial Supervisor)	Saad Kamar
Contact Number(s)	+201122298785

Internship/Industrial Placement Report clearance by Company:

Signature	Company Stamp	Date
		15/5/2026

Note: If the company wishes to have a copy of the report, the arrangement is left between the company and the student.

Internship/Industrial Placement Assessment Form
(To be filled by Industrial Supervisor)



Instruction
This form is to be completed by Industrial Supervisor. The completed Assessment form must be submitted together with Internship Report.

General Information	
Student Name	Adam Mohamed Bannani
Award Title	Product Designer
Name of Employing Company	Caliber
Name of Industrial Supervisor	Saad kamar
Internship Start Date	15/1/2026
Internship End Date	15/5/2026

Student's Performance		
Please rate the student's performance using the following: A = Excellent B = Good C = Satisfactory D = Poor U = Untested		
	Rating	Comments
Attitude to supervision	C	He is not very obedient when an order feels like an order.
Social integration	A	Can deal with all types of people.
motivation	A	He showed strong hunger to finish more projects which was impressive to me.
Perseverance	A	He fails and he tries again till he gets it the way he wants it but crazy determination
Technical knowledge	A	
Productivity	A	
Capacity for teamwork	B	He likes to always lead which is good but not in all situations
Problem solving ability	B	Very very good for his age, always searches for new solutions
Communication skills – written	B	
Communication skills – oral	A	
Others (please specify)		

Note: The rating will be used only as feedback and not for grading

Overall Grade					
The mark awarded will be used only as a feedback and not for grading. Please tick ONE box.					
Unsatisfactory	Weak	Satisfactory	Good	Very Good	Exceptional
☐	☐ ☐	☐ ☐	☐ ☐	☐ ☐	☒

1	2	3	4	5	6	7	8	9	10
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Comments
Overall he was an extremely good addition to the team. He showed a lot of determination and impressed me on how he deals and leads workers. One thing that is maybe negative about him that I would like to comment about is that he was a bit egotistic which is not very bad but if you are working under a lot of people you have to let go of your own ego sometimes. But other than that, he showed very good communication and design skills which impressed me a lot for someone his age. His design are extremely unique but sometimes he struggles to design something that looks very normal because he's always searching for the best design possible and how to improve it and sometimes a designer has to be able to design ordinary thing that looks ordinary.

Industrial Supervisor's signature & stamp 	26/5/2026 Date:
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INTERNSHIP/INDUSTRIAL PLACEMENT STUDENT FEEDBACK FORM



Please fill in the form diligently and ensure that it is **attached at the back of your Internship/Industrial Placement Report for submission.**

General Information	
Student Name	Adam Mohamed Bannani
Student ID No	TP073524
Intake (when the placement starts)	January 2026
Student Passport / IC No	A28421283
School	Industrial design
Period of Placement (Start Date and End Date)	!5/1/2026 – 15/5/2026
APU Academic Supervisor	Jack Lai Yong Geat

Feedback on Company	
Company Name	Caliber
Company Address	6 th of October, Cairo, Egypt
Company Supervisor & Contact no/email address	+201122298785. saadkamar716@gmail.com
Department that Student was Attached to	Design department/ product design

Briefly describe the nature of work performed:

Producing technical drawings for all assigned products and projects

Handling customized projects from start to finish, including:

Design and detailed drawings

Coordination and alignment with the workshop

Time management of production

Overseeing samples and revisions

Following through until the final approved product

Preparing and pricing customized products in coordination with the factory/workshop, in- cluding material selection, production methods, and cost breakdowns, ot support accurate project quotations.

Quality control oversight during production to ensure designs are executed as intended. Designing new collections, from concept development to production-ready designs.

Was the nature of work assigned to you challenging?	Yes
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If not, please elaborate:

Work was challenging because I had to finish designing furniture items and execute it all at one which mean I was responsible from designing to manufacturing. It was more challenging because I worker in the B2B section, so a lot of items had to be executed from design to manufacturing in a short time because the faster I finished the projects(orders) the more money they gave so I would say it was challenging in that sense.

Briefly comment on the level of supervision provided:

Level of supervision was overall good. I had all the support I need if I wanted to understand more or ask about something.	
Would you recommend that we continue to assign students to this company in future?	Yes
Suggest ways in which you think the Industrial Placement Programme offered by the company could be improved: I think university should offer more knowledge about executing the manufacturing process not just taking lectures or studying about it because product design and manufacturing in general is a very handy job. Also I am a person that likes to work more with metal and in university workshop, they do not provide us any metal related equipment because most of the equipment is for woodworking which made me explore a lot outside of university in order to physically learn these things, maybe they should consider bring more equipment related to other materials than wood, like metal and glass for example.	
Overall, how do you rate the company in providing you with this training?	Good
If poor, please elaborate: The only problem I had with the company is that there was no strict systematic systems on manufacturing so I offered them ways to improve the manufacturing cycles so things can be more efficient and get done faster.	
Allowance Paid	About 5000-7000 rm

General Feedback (optional)
Suggest ways in which you think the Industrial Placement Program in APU that can be improved: I think university should offer more knowledge about executing the manufacturing process not just taking lectures or studying about it because product design and manufacturing in general is a very handy job. Also I am a person that likes to work more with metal and in university workshop, they do not provide us any metal related equipment because most of the equipment is for woodworking which made me explore a lot outside of university in order to physically learn these things, maybe they should consider bring more equipment related to other materials than wood, like metal and glass for example.

Date: 29/5/2026

Signature of Student:



Internship/Industrial Placement Visit Report

(To be filled by (Mentor) Academic Supervisor)



Instruction
The visit report is to be completed by Academic Supervisor after each site visit. If an in-person site visit is not feasible because of geography or other reasons, the Academic Supervisor and student should arrange a virtual site visit using appropriate technology. Only one site visit is required unless there is any special request for more visits by the University or the host company. The completed visit report should be submitted together with Internship Report.

General Information	
Student Name	Adam Mohamed Bannani
Academic Supervisor	Jack Lai Yong Geat
Award	BA (Hons) in Industrial Design
Company	Caliber
Industrial Supervisor	Jack Lai
Date of Visit	4 June 2026
Number of Visit	☒ One ☐ Two ☐ More than two times
Type of Communication	☐ Visit ☒ Email ☐ Telephone Call ☐ Video Conferencing

Please tick 'Yes' / 'No'			
	Yes	No	Comments
An introductory briefing session has been conducted for the student	V		
The student reported for duty on the expected date	V		
The student has been punctual	V		
The student has displayed good work ethics (performance)	V		
The student has displayed willingness to learn	V		
The student understands the tasks allocated	V		

Summary	Yes	No	Comments
Is the performance of the student satisfactory?	V		
Is the experience provided by the company adequate?	V		
Do the discussion notes below require prompt attention by the Internship Coordinator?	V		

Notes of Discussion

Please describe the overall performance of students and the work environment, including student's feedback and industrial supervisor's feedback. Please specify the area(s) that requires immediate attention.

Adam has performed very well throughout his internship and has met my expectations as an intern. He demonstrated a level of confidence and practical experience that is uncommon for students, particularly in factory environments. He was comfortable visiting factories independently, communicating with workers, and handling manufacturing-related discussions professionally.

His CAD modelling skills were also impressive. I frequently assigned him modelling tasks, and he consistently completed them accurately and efficiently, often within a very short timeframe.

Adam showed strong initiative, problem-solving ability, and a willingness to take responsibility for his work. He adapted quickly to the demands of the job and became a valuable contributor to the team.

One area for development would be adapting to different workplace structures and schedules. Adam preferred managing tasks according to his own workflow, which he felt improved his productivity. While this often produced good results, developing greater flexibility in working within established organizational processes would benefit him further.

As a suggestion for the university, I believe students would benefit from more factory visits and direct industry exposure. Many practical skills required in manufacturing and product development are learned through real-world experience rather than in the classroom.

Overall, Adam left a very positive impression on our team. Based on his performance, I would be happy to consider him for future employment opportunities, and I have already expressed my interest in having him return after completing his studies.

Mentor's signature:
Jack Lai Yong Geat



05 June 2026

Date:

Internship Attendance Record

Industrial Supervisor Name: Saad Kamar	Student Name: Adam Mohamed Bannani
Company Name: Caliber	Student ID: TP073524
Company Address: 6 th of October, Cairo, Egypt	Student IC: A28421283

Week 1

Workplace	Day	Date	Time In	Time Out	Time Served
Factory	thursday	19/3	9 am	3 pm	6 hours
-	Sunday	22/3	-	-	Holiday (Eid Al-Fitr)
-	Monday	23/3	-	-	Holiday (Eid Al-Fitr)
-	Tuesday	24/3	-	-	Holiday (Eid Al-Fitr)
Online	wednesday	25/3			3-5 hours

Week 2

Workplace	Day	Date	Time In	Time Out	Time Served
Factory	thursday	26/3	9 am	3 pm	6 hours
- Online	Sunday	29/3			3-5 hours
factory	Monday	30/3	9 am	5 pm	8 hours
factory	Tuesday	31/3	9 am	5 pm	8 hours
Online	wednesday	1/4			3-5 hours




Industrial Supervisor's Signature & Stamp

Date: 1/ 4 / 2026

Internship Attendance Record

Week 3

Workplace	Day	Date	Time In	Time Out	Time Served
Factory	thursday	2/4	9 am	3 pm	6 hours
Online	Sunday	5/4			3-5 hours
factory	Monday	6/4	9 am	5 pm	8 hours
factory	Tuesday	7/4	9 am	5 pm	8 hours
Online	wednesday	8/4			3-5 hours

Week 4

Workplace	Day	Date	Time In	Time Out	Time Served
Factory	thursday	9/4	9 am	3 pm	6 hours
Online	Sunday	12/4			3-5 hours
-	Monday	13/4	-	-	Holiday (Sham El- Nessim)
factory	Tuesday	14/4	9 am	5 pm	8 hours
Online	wednesday	15/4			3-5 hours

Week 5

Workplace	Day	Date	Time In	Time Out	Time Served
Factory	thursday	16/4	9 am	3 pm	6 hours
Online	Sunday	19/4			3-5 hours
factory	Monday	20/4	9 am	5 pm	8 hours
factory	Tuesday	21/4	9 am	5 pm	8 hours
Online	wednesday	22/4			3-5 hours



Industrial Supervisor's Signature & Stamp

Date: 22/ 4 / 2026

Internship Attendance Record

Week 6

Workplace	Day	Date	Time In	Time Out	Time Served
Factory	thursday	23/4	9 am	3 pm	6 hours
Online	Sunday	26/4			3-5 hours
factory	Monday	27/4	9 am	5 pm	8 hours
factory	Tuesday	28/4	9 am	5 pm	8 hours
Online	wednesday	29/4			3-5 hours

Week 7

Workplace	Day	Date	Time In	Time Out	Time Served
Factory	thursday	30/4	9 am	3 pm	6 hours
Online	Sunday	3/5			3-5 hours
factory	Monday	4/5	9 am	5 pm	8 hours
factory	Tuesday	5/5	9 am	5 pm	8 hours
Online	wednesday	6/5			3-5 hours

Week 8

Workplace	Day	Date	Time In	Time Out	Time Served
Factory	thursday	7/5	9 am	3 pm	6 hours
Online	Sunday	10/5			3-5 hours
factory	Monday	11/5	9 am	5 pm	8 hours
factory	Tuesday	12/5	9 am	5 pm	8 hours
Online	wednesday	13/5			3-5 hours



Industrial Supervisor's Signature & Stam

Date: 13/ 5 / 2026

Internship Attendance Record

Week 9

Workplace	Day	Date	Time In	Time Out	Time Served
Factory	thursday	14/5	9 am	3 pm	6 hours



Industrial Supervisor's Signature & Stamp

Date: 14/ 5 / 2026



ASIA PACIFIC UNIVERSITY OF TECHNOLOGY & INNOVATION

School of Technology

IT001-4-2 & IT001-4-3
INTERNSHIP / INDUSTRIAL TRAINING
REPORT

Student Name	Adam Mohamed Bannani
Student ID	TP073524
Intake	March 2024
Programme	Bachelor of Science (Hons) in Industrial Design
Company	Caliber
Internship Period	15 January 2026 – 15 May 2026
Academic Supervisor	Jack Lai Yong Geat

Acknowledgements

I would like to express my sincere gratitude to everyone who supported me throughout my 17-week internship at Caliber, 6th of October, Cairo, Egypt.

First and foremost, I wish to thank my Industrial Supervisor, Mr. Saad Kamar, for his guidance, trust, and invaluable mentorship. His willingness to involve me in real client projects from the very first week, and to challenge me with responsibilities beyond what is typically expected of an intern was instrumental in shaping me as a designer and as a professional. I am particularly grateful for the honest feedback he provided, which helped me grow both technically and personally.

I would also like to thank my Academic Supervisor, Mr. Jack Lai Yong Geat, for his continued support, timely communications, and constructive advice throughout the internship period. His guidance during supervisory sessions kept me on track and focused on producing quality documentation alongside my practical work.

My gratitude also extends to the entire team at Caliber — the factory workers, blacksmiths, and workshop staff — who were always willing to share their craft knowledge and collaborate with me on the production floor. This internship would not have been as enriching without their patience and expertise.

Finally, I am grateful to Asia Pacific University of Technology & Innovation (APU) for providing this structured internship programme, which has been an invaluable bridge between academic learning and real-world professional practice.

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SECTION 1: COMPANY INTRODUCTION

1.1 Introduction to the Business

Caliber is a bespoke furniture design and manufacturing company based in the 6th of October City, Cairo, Egypt. The company specialises in the design and production of high-end, custom-made furniture and interior pieces, serving both individual clients and business-to-business (B2B) partners. Caliber occupies a niche within the Egyptian furniture market by offering a fully integrated service model: from initial concept and 3D design through to in-house fabrication, finishing, and delivery.

The company operates across two primary environments — a design office where product concepts are developed and technical drawings are produced, and a factory/workshop where the physical manufacturing takes place. This dual-environment structure allows Caliber to maintain tight quality control over its output while offering a high degree of customisation to clients.

I was placed at Caliber as a Product Designer intern, working across both the design and production sides of the business. My role involved end-to-end ownership of projects: from 3D modelling and technical drawing production to overseeing the manufacturing of the finished pieces on the factory floor.

1.2 The Organisation

Caliber was established as a small-to-medium enterprise (SME) operating in the premium custom furniture segment. Its client base includes interior designers, renovation studios, hospitality venues, and private residential clients. The company's competitive advantage lies in its ability to translate unique client briefs into fully custom, production-ready furniture pieces using a combination of advanced digital design tools and skilled craftsmanship.

The organisational structure of Caliber is relatively flat. Mr. Saad Kamar, who served as my Industrial Supervisor, oversees both the design and production operations. Under his leadership, a small team of designers, factory workers, blacksmiths, and finishing specialists collaborate to execute each project. This lean structure meant that as an intern, I was given direct access to decision-makers and was expected to operate with a significant degree of independence from an early stage.

The company's products span a wide range of furniture typologies including chairs, tables, benches, shelving, lighting, and custom installation pieces. Materials commonly used include stainless steel, sheet metal, iron, chrome-plated tube, solid wood, MDF, glass, resin, and a variety of upholstery fabrics and leathers.

1.3 The Environment

Caliber operates within the Egyptian luxury and semi-luxury furniture market, a sector that has experienced growing demand in recent years driven by an expanding real estate sector and a rising number of hospitality and commercial interior projects. The 6th of October City, where Caliber is located, is a major industrial and commercial hub west of Cairo, making it strategically positioned for manufacturing operations.

The competitive landscape in this market includes both local manufacturers and international brands. Caliber differentiates itself through its B2B model, customisation capability, and the integration of digital design tools (CAD/3D modelling) into its production workflow — a feature that is not universally adopted by competitors in the Egyptian market.

Technology plays a central role in Caliber's operations. The adoption of software such as Fusion 360, Shapr3D, and SolidWorks, combined with laser cutting technology and CNC-adjacent processes, positions Caliber at the more technically advanced end of the local furniture manufacturing spectrum. One challenge identified during the internship was the absence of a fully systematised manufacturing process, which occasionally led to inefficiencies on the production floor. I proactively proposed process improvements to help streamline production cycles.

1.4 The Functions

Caliber's operations are organised around two core functional areas:

Design & Engineering: This function is responsible for conceptualising products, producing 3D models, generating technical drawings, preparing DXF files for laser cutting, and managing client briefs. As a Product Designer intern, I was embedded within this function.

Manufacturing & Production: This function encompasses all physical fabrication activities including laser cutting, tube bending, welding, blacksmithing, powder coating, painting, upholstery, and final assembly. The factory operates on a project-by-project basis, with production timelines driven by client order deadlines.

A third, more informal function is sales and client management, which involves taking client briefs, producing quotations, and managing delivery. I was exposed to aspects of this function through my involvement in B2B order management, material sourcing, and client-driven customisation projects.

1.5 The Department / Section

I was attached to the Design Department / Product Design section of Caliber. Within this department, my responsibilities covered the full design-to-production pipeline for

assigned projects. On any given project, my workflow would typically begin with receiving a client brief or supervisor instruction, followed by conceptual development, 3D modelling, technical drawing production, DXF file preparation, and handoff to the manufacturing team — after which I would move onto the factory floor to supervise and quality-check the execution.

This holistic involvement across the entire product lifecycle — from digital design to physical realisation — was a defining feature of my internship and provided me with a depth of understanding that purely design-focused or purely production-focused roles would not have offered.

1.6 Communications and Information Technology

Communication within Caliber was largely informal and direct, reflecting the company's flat organisational structure. Day-to-day project communication between the design and production teams was handled in person, with supplementary communication via WhatsApp messaging — including the sharing of technical drawings and DXF files with factory supervisors and workshop partners.

On the technology side, the design workflow was built around industry-standard CAD and 3D modelling software: Fusion 360, Shapr3D, and SolidWorks. These tools were used to develop 3D models, extract technical drawings with full dimensional annotation, and generate DXF files for laser cutting. The use of digital files significantly improved the accuracy and speed of manufacturing compared to hand-drawn or loosely specified briefs.

One area where I observed a gap was in the absence of a formal project management or workflow tracking system. Communication was effective on a team level but lacked a structured record-keeping mechanism, which I identified as an area for potential process improvement.

1.7 Financial Structure

As a privately held SME, Caliber's detailed financial information was not disclosed during the internship. However, from my involvement in B2B order management and project quotation activities, I was able to observe several aspects of the company's financial operations.

Revenue is generated primarily through client orders, with pricing based on material costs, production complexity, and delivery timelines. B2B clients typically place bulk orders (e.g., 9 coffee tables, 16 chairs, 14 dining chairs), which represent significant revenue events for the company. I was involved in preparing and pricing customised products in coordination with the factory and workshop, including material selection, production methods, and cost breakdowns to support accurate project quotations.

The company's cost structure is driven by raw material procurement (stainless steel, sheet metal, chrome tube, fabrics, glass), labour costs across the design and production teams, and outsourced specialist services (e.g., resin mould workshops and external fabricators for specific processes).

1.8 Introduction to the Nature of Work

My internship at Caliber spanned 16 weeks from 15 January 2026 to 15 May 2026. The nature of work was project-driven, with each week typically centred on one or more active client orders. I was responsible for managing assigned projects from initial design concept through to final quality-checked delivery.

Key work activities throughout the internship included:

- 3D modelling of furniture pieces using Fusion 360, Shapr3D, and SolidWorks
- Production of detailed technical drawings with full dimensional annotation
- Generation of DXF files for laser cutting sheet metal and tube components
- B2B order management: translating client requirements into manufacturable designs
- Inventory checking and material sourcing
- Factory floor supervision during manufacturing
- Quality control checks prior to product delivery
- Material and finish specification (metals, fabrics, leathers, coatings)
- Coordination with external workshops for specialist processes
- Freelance furniture collection design for an external client studio
- Process improvement recommendations for the manufacturing workflow

1.9 Tasks – Application of Concepts and Knowledge

Throughout the internship, I was able to directly apply theoretical knowledge gained during my academic studies in Industrial Design to real-world professional challenges. The following highlights the most significant projects undertaken and the academic concepts applied:

Weeks 1–2: Orientation and Initial Design Work

During the first two weeks, I familiarised myself with the company's design and production workflow. I began producing 3D models and technical drawings using Fusion 360, Shapr3D, and SolidWorks — applying CAD skills developed in university coursework. I also prepared DXF files for laser cutting, directly applying knowledge of manufacturing file formats and tolerances. My first B2B order involvement required me to translate loosely defined client requirements into precise, manufacturable designs, applying design communication and technical drawing principles.

Weeks 3–4: B2B Order Management and Re-engineering

In Week 3, I managed a full B2B order from brief to production handoff, conducting inventory checks and coordinating with the production team to ensure on-time delivery. In Week 4, I re-engineered existing furniture items into new 3D models with updated technical drawings — applying reverse-engineering principles and critical design thinking to improve upon existing products.

Week 5: Stainless Steel Coffee Table Production (9 Units)

I designed and oversaw the production of 9 stainless steel coffee tables for a B2B client. This project applied my knowledge of sheet metal design, structural integrity, laser cutting processes, and quality assurance. The pre-polishing stage and final QC check required an understanding of surface finish standards and manufacturing tolerances.

Week 6: Lounge Chair and Buffet – Complex Manufacturing

Working on a lounge chair that required precise bending and cutting, I visited the factory to oversee the process and ensure accuracy. This week introduced me to additional machinery on the factory floor, expanding my practical manufacturing knowledge. The project applied ergonomics knowledge (chair dimensions, proportions) alongside manufacturing supervision.

Week 7: Material Sourcing and Innovation

My supervisor tasked me with sourcing materials and manufacturers for a glass-top table concept. Recognising that solid glass would be impractically heavy, I proposed using resin as an alternative material — applying problem-solving and materials science knowledge. I successfully identified an external workshop capable of producing large resin moulds, demonstrating initiative and supply chain awareness.

Week 8: Wooden Chair B2B Order (11 Units)

I re-engineered a coffee table design and oversaw a B2B order of 11 laser-cut wooden chairs assembled using a layered lamination technique. This project applied my understanding of wood manufacturing processes, laser cutting design for CNC production, and quality control for batch orders.

Week 9: Fabric Application and Surface Finishing (16 Chairs)

Working on the lounge chair designed in Week 4, I selected appropriate leather, oversaw the welding gap-filling and powder-coat painting process, and developed a wooden backrest substrate to allow fabric upholstery on a metal frame. This project significantly expanded my knowledge of multi-material finishing processes and upholstery techniques.

Part 2 – Week 1: Round Glass-Top Coffee Tables (5 Units)

I 3D-modelled a chrome tubular base for 5 round coffee tables in Shapr3D, engineering the arm geometry to ensure structural integrity. I supervised tube bending, welding, polishing, and glass mounting, performing final QC on all units before delivery.

Part 2 – Week 2: Two Chair Designs (7 + 14 Units)

I modelled and coordinated production for two chair designs simultaneously — a compact lounge chair and a dining/accent chair — demonstrating the ability to manage multiple concurrent projects and apply different structural and upholstery design principles to each piece.

Part 2 – Week 3: Technical Drawing Production

I produced a complete set of manufacturing-standard technical drawings for an ongoing furniture order, covering multiple components including panels, structural elements, and seating base frames with full dimensional annotation.

Part 2 – Week 4: Book Stand Design and Prototype

I designed a geometric sheet metal book stand from concept through to a first physical prototype. The prototype was fabricated in raw steel to validate the geometry and proportions before full production — applying prototype testing methodology and iterative design principles.

Part 2 – Week 5: Ceiling-Suspended DJ Table

One of the most technically demanding projects of the internship: a stainless steel DJ table suspended from a ceiling via steel rods and chains. I modelled the platform in SolidWorks, integrating the suspension system into the structural design, coordinated fabrication, and supervised on-site installation. This project applied structural engineering principles alongside interior design considerations.

Part 2 – Weeks 6–7: Freelance Furniture Collection Design

I designed a complete cohesive furniture collection for an external creative studio — including a lounge chair, bench, side table, coffee table, cylindrical stool, and floor lamp — sharing a teal/chrome design language. In the second phase, I refined the collection by introducing an arch motif inspired by Islamic and Moroccan architectural forms, producing final high-quality renders for client presentation.

Part 2 – Week 8: Custom Curved Iron Chair (4 Units)

I designed a classically inspired armchair with a sculptural scrolled X-frame base in Shapr3D, modelling complex tube bending paths to ensure geometrical consistency and manufacturability. Four units were produced under my supervision.

Part 2 – Week 9: 3D Modelling for Upcoming Production

In the final week, I modelled a mid-century lounge chair with an angular wooden X-frame base and a dining table with a tapered pyramid base — both prepared to technical drawing extraction standard and handed off to production as a deliverable for future orders.

1.10 Development of Employability Skills

The 16-week internship at Caliber was a transformative experience that significantly accelerated my development across a wide range of employability skills. The following outlines the key areas of growth:

Technical Design Skills

My proficiency in 3D modelling software — particularly Shapr3D, Fusion 360, and SolidWorks — was elevated substantially through daily professional application. The speed and accuracy with which I could produce models and technical drawings improved week-on-week, and I developed a much more nuanced understanding of design-for-manufacture constraints. My supervisor rated my technical knowledge as 'A' (Excellent) in his end-of-internship assessment.

Manufacturing Knowledge

Working on the factory floor exposed me to manufacturing processes that are rarely taught practically in university: tube bending, welding, powder coating, laser cutting file preparation, layered wood lamination, resin mould sourcing, and fabric upholstery on metal frames. This hands-on exposure represents a significant addition to my skill set as a product designer.

Project Management and Order Coordination

Managing B2B orders end-to-end — from brief interpretation through design, inventory checking, production coordination, and quality control — developed my project management capabilities considerably. I learned how to balance design quality with time and resource constraints in a real commercial environment.

Problem-Solving and Initiative

Throughout the internship I was consistently placed in situations that required creative problem-solving: proposing resin as an alternative to glass, developing wooden substrates for fabric application on metal, suggesting process improvements to the manufacturing workflow, and sourcing specialist external workshops. My supervisor noted: 'Very very good for his age, always searches for new solutions.'

Communication and Leadership

Working directly with factory workers, blacksmiths, and external workshop partners required clear, effective communication of technical requirements. I learned to convey complex design intentions to non-design professionals in practical, actionable terms. My supervisor noted my strong oral communication skills (rated 'A') while also identifying that I tend to take a leadership role — a strength that I am learning to balance with collaborative flexibility.

Professional Conduct

Operating within a professional environment with real client deliverables and financial consequences developed my understanding of workplace standards, timekeeping, confidentiality, and professional accountability. The experience of receiving frank supervisor feedback — including the observation that I can be less receptive to direct instructions — has provided me with important self-awareness for my professional development.

SECTION 2: FUTURE PLANS FOR EMPLOYMENT

2.1 Updated Plan for Employment and Continuing Professional Development

The internship at Caliber has significantly refined and focused my career aspirations. Prior to the placement, my career goals were broadly oriented towards product design; the 16 weeks at Caliber have crystallised this into a specific ambition to work at the intersection of furniture design, metal fabrication, and high-end interior product development.

In the short term (0–2 years), my goal is to complete my degree at APU and secure a junior product designer position within a design studio or furniture manufacturer that operates at the premium/custom segment of the market. The experience gained at Caliber — particularly in B2B project management, 3D modelling for manufacture, and factory supervision — has given me a strong and differentiated portfolio that I will continue to build and refine. I intend to develop my skills in product visualisation (rendering and photography) to better communicate design proposals to clients, and to deepen my proficiency in SolidWorks, which I identified as the most powerful tool for complex structural design during the internship.

In the medium term (2–5 years), I aim to progress towards a senior designer or design lead role, taking greater responsibility for creative direction on projects. Given my interest in materials and manufacturing, I would also like to pursue continuing professional development in advanced manufacturing techniques — particularly CNC machining, metal casting, and advanced composite materials — to expand the range of design solutions I can offer. Courses and certifications from industry bodies such as the Chartered Society of Designers (CSD) would support this development.

In the longer term, my ambition is to establish my own design practice or furniture brand, drawing on the end-to-end production knowledge gained at Caliber. The ability to design, engineer, and manufacture — rather than just design — is a competitive advantage that I am committed to developing throughout my career.

2.2 Plan for Securing Good Quality Internships for Future Students

Based on my internship experience at Caliber, the following recommendations are offered to assist future students in securing high-quality industrial placements in product design and related fields:

Build a Practical Portfolio Before Applying

Many design companies — particularly in the manufacturing sector — prioritise demonstrable practical skills over academic credentials alone. Future students should develop a portfolio that includes 3D models, technical drawings, and rendered concepts

produced using industry-standard tools (Fusion 360, SolidWorks, Shapr3D, Rhino). Employers like Caliber respond strongly to applicants who can show they can 'hit the ground running.'

Target Manufacturing-Integrated Design Companies

For product design students, placements at companies that combine design with in-house manufacturing provide far richer learning experiences than purely studio-based environments. Seek out furniture manufacturers, metal fabricators, and custom interior product companies that have both design and production capabilities.

Communicate Proactively with the Industrial Supervisor

From the first week, establish clear communication with the Industrial Supervisor about expectations, workload, and the types of projects you would like to be involved in. Supervisors appreciate interns who show initiative and actively seek out opportunities rather than waiting to be assigned work.

Embrace the Factory Floor

One of the most valuable aspects of my internship was time spent on the factory floor. Future students should not limit themselves to the design office — understanding how designs are physically realised is essential for becoming a well-rounded product designer.

Document Everything Consistently

Maintaining a thorough and up-to-date logbook from the first week makes the final report submission significantly easier. Photograph your work regularly, note down key decisions and lessons learned, and keep copies of all technical drawings produced.

2.3 Conclusions and Recommendations

The 16-week internship at Caliber has been one of the most formative experiences of my academic career. Working as a Product Designer in a real commercial environment — managing client orders, producing professional technical documentation, supervising manufacturing, and delivering finished products — has provided a depth and breadth of practical education that complements and in many ways surpasses what can be achieved in a university setting alone.

The key conclusions drawn from this experience are as follows:

- Design and manufacturing are inseparable disciplines. A product designer who understands manufacturing processes at a practical level produces better, more manufacturable designs.
- Soft skills are as important as technical skills. The ability to communicate complex design information to non-designers, to manage time under commercial pressure, and to accept and act on critical feedback are all essential professional competencies.

- Initiative and problem-solving are highly valued in professional settings. The projects where I added the most value were those where I identified a problem and proposed a solution proactively — such as the resin material substitution and the process improvement recommendations.
- Self-awareness is a critical component of professional development. The feedback from my Industrial Supervisor — particularly regarding attitude to supervision and ego management — has given me valuable self-awareness that I will carry forward into my professional career.

Recommendations for APU:

- Introduce more hands-on manufacturing workshops, particularly in metalworking (sheet metal, welding, tube bending). The current university workshop provision is predominantly wood-focused, which limits the preparation of product design students for industries that work with metal and glass.
- Integrate design-for-manufacture principles more deeply into the curriculum, with real constraints (material cost, production time, batch quantities) included in project briefs.
- Encourage students to develop proficiency in multiple CAD platforms (particularly SolidWorks and Fusion 360) before entering the workplace, as these are the most commonly required tools in product and furniture design companies.

Overall, I would strongly recommend Caliber as a placement company for future APU product design students. The breadth of project exposure, the quality of practical learning, and the professional culture at Caliber make it an exceptional environment for an industrial design intern to develop both technical and professional skills.

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Internship Logbook

Instruction:
Student is required to record the daily internship activities, learning and experience using this logbook. At the end of each week, the Industrial Supervisor needs to review the logbook and verify it. The completed logbooks must be submitted together with the internship reports.

Student Name	Adam Mohamed Bannani
Student ID	TP073524
Company Name	Caliber

Week: 1	Start Date: 22/3/2026	End Date: 26/3/2026
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3D modelling and manufacturing of 5 round glass-top coffee tables.
The objective was to design the tubular chrome base in Shapr3D, produce technical drawings, and oversee the full manufacturing process from tube bending and welding to final assembly and quality check.

This week, I was tasked with designing and overseeing the production of 5 round coffee tables featuring a smoked glass top and a polished chrome tubular base.

I began by 3D modelling the base structure in Shapr3D, carefully engineering the arm geometry and dimensions — with each arm extending approximately 52.5 cm from the central column — to ensure structural integrity and visual balance. Once the model was finalised, I produced technical drawings and DXF files to guide the factory team.

In the factory, I supervised the tube bending, welding, and assembly of the chrome bases. After polishing, the smoked round glass tops were mounted onto the frames. I carried out quality checks on all 5 units before they were cleared for delivery, ensuring dimensional accuracy and finish consistency throughout the batch.



Industrial Supervisor's signature & stamp:

Date: 26/3/2026



Internship Logbook

Instruction:
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Student Name	Adam Mohamed Bannani
Student ID	TP073524
Company Name	Caliber

Week: 2	Start Date: 29/3/2026	End Date: 2/4/2026
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3D modelling and manufacturing coordination of two chair designs.

The objective was to model both chairs, produce technical drawings, and hand off to the manufacturing team for production.

This week, I was responsible for 3D modelling two different chair designs and coordinating their production with the manufacturing team.

The first design was a compact lounge chair with a polished chrome rectangular frame, a cylindrical bolster backrest, and a fully upholstered seat in a dark brown/chocolate fabric. I modelled the frame geometry and cushion profiles, then produced the technical drawings required for fabrication. 7 units of this chair were manufactured.

The second design was a lighter dining/accent chair with angled tubular metal legs, padded seat and backrest, and integrated armrests — upholstered in a beige/cream fabric. After finalising the 3D model and technical drawings, I passed the files to the manufacturing team. 14 units of this chair were produced.

Both models were completed and handed off by the end of the week, with all dimensional and structural details clearly communicated through the technical drawings.



Industrial Supervisor's signature & stamp:



Date: 2/4/2026

Internship Logbook

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Student Name	Adam Mohamed Bannani
Student ID	TP073524
Company Name	Caliber

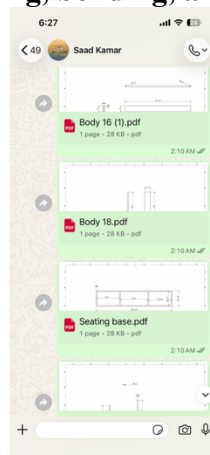
Week: 3	Start Date: 5/4/2026	End Date: 9/4/2026
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Technical drawing production for a furniture order.
The objective was to produce precise technical drawings for individual furniture components as requested by the supervisor, to guide the manufacturing team in cutting, bending, and assembling the parts.

This week, my supervisor assigned me to produce a set of technical drawings needed for an ongoing furniture production order. The drawings covered multiple individual components of the furniture piece, each with fully annotated dimensions.

The components I drew included flat rectangular panels, vertical structural elements, and a seating base frame — the latter spanning approximately 255 mm in total width, subdivided into precise internal sections, with detailed height and thickness annotations.

Each drawing was prepared to manufacturing standard, ensuring the factory team had all the dimensional data required for accurate laser cutting, bending, and assembly. All drawings



were delivered to the supervisor upon completion.

Industrial Supervisor's signature & stamp: Date: 9/4/2026

Internship Logbook

Instruction:
Student is required to record the daily internship activities, learning and experience using this logbook. At the end of each week, the Industrial Supervisor needs to review the logbook and verify it. The completed logbooks must be submitted together with the internship reports.

Student Name	Adam Mohamed Bannani
Student ID	TP073524
Company Name	Caliber

Week: 4	Start Date: 12/4/2026	End Date: 16/4/2026
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Design, technical drawing, and prototype manufacturing of a book stand.
The objective was to design a geometric sheet metal book stand, produce the required drawings, and fabricate a first sample to validate the structural design and dimensions before moving to full production.

This week, I was tasked with designing a book stand from scratch. The design features an angular geometric base constructed from flat sheet metal panels arranged in a triangular/pyramid-like structure, converging at the top to support a flat inclined reading platform.

I developed the 3D model and produced technical drawings with all required dimensions, then passed the DXF cutting files to the factory team. The first sample was fabricated in raw steel to physically validate the geometry, structural stability, and proportions of the design before committing to full production.

The prototype was reviewed and confirmed the design concept. Any necessary adjustments to dimensions or assembly details were noted for the final production run.



Industrial Supervisor's signature & stamp:



Date: 16/4/2026

Internship Logbook

Instruction:
Student is required to record the daily internship activities, learning and experience using this logbook. At the end of each week, the Industrial Supervisor needs to review the logbook and verify it. The completed logbooks must be submitted together with the internship reports.

Student Name	Adam Mohamed Bannani
Student ID	TP073524
Company Name	Caliber

Week: 5	Start Date: 19/4/2026	End Date: 23/4/2026
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3D modelling and manufacturing of a ceiling-suspended DJ table. The objective was to design a DJ booth platform that mounts to the ceiling via suspended chains and steel rods, fabricate it in stainless steel, and oversee its installation in a modern venue.

This week, I was assigned to design and oversee the production of a DJ table that is suspended from the ceiling rather than floor-mounted — an unconventional piece combining structural engineering with interior design.

I began by 3D modelling the table in SolidWorks, designing a large stainless steel platform with a raised secondary shelf at one end to accommodate DJ equipment. The suspension system was integrated into the model, using vertical steel rods and chains that anchor to the ceiling structure above.
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The platform was fabricated in the factory from stainless steel sheet, with edges folded and welded to achieve the clean box-section finish visible in the final piece. After fabrication, I coordinated the on-site installation, ensuring the table was level, securely suspended, and correctly positioned within the venue. The final result was installed in a modern industrial-style space with exposed concrete walls and ceilings.
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Industrial Supervisor's signature & stamp:

Date: 23/4/2026



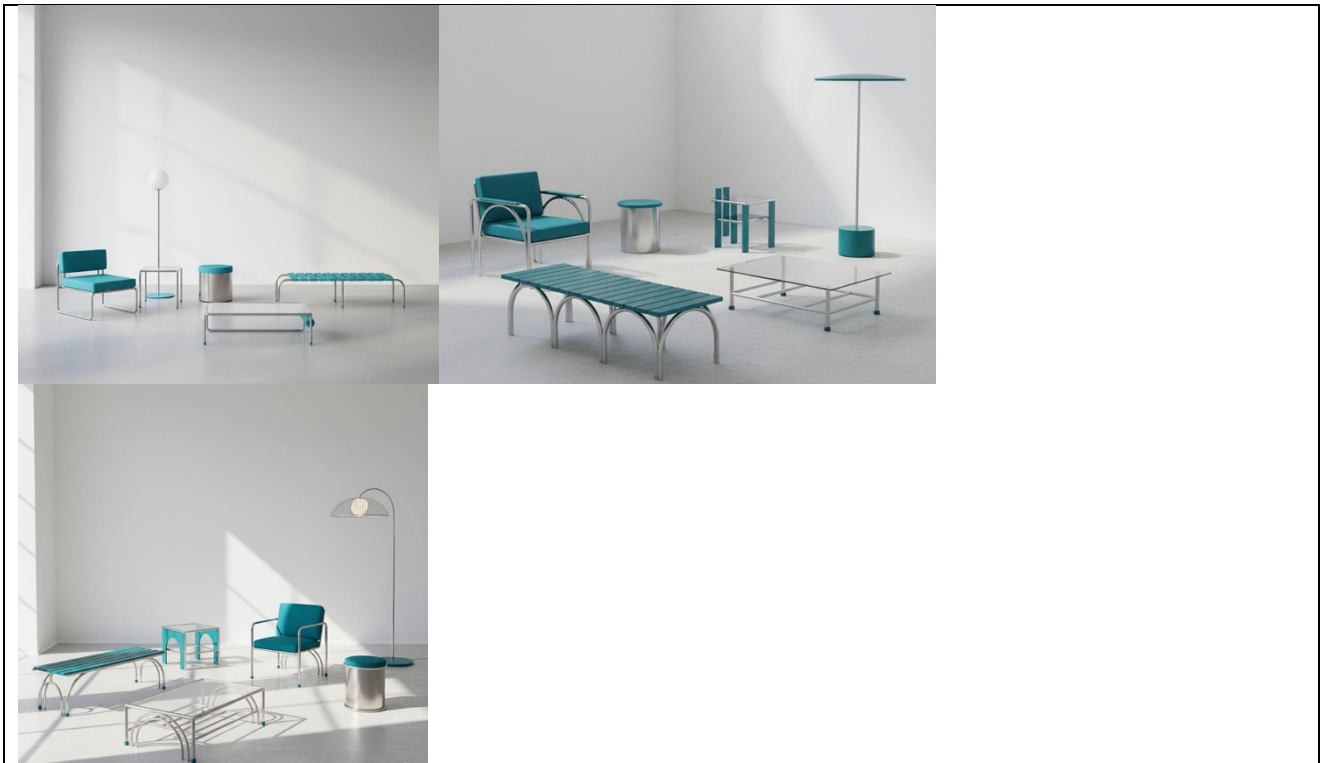
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Internship Logbook

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Student Name	Adam Mohamed Bannani
Student ID	TP073524
Company Name	Caliber

Week: 6	Start Date: 26/4/2026	End Date: 30/4/2026
Freelance furniture collection design — Phase 1 (concept & initial modelling). The objective was to design a cohesive furniture collection for a creative studio that does renovation projects and wanted to launch their own furniture line, ensuring all pieces aligned with their brand identity. This week marked the beginning of a two-week freelance project for a creative studio. The studio, which specialises in renovation projects, commissioned a full furniture collection that would reflect their brand identity and serve as the foundation of their own furniture line. I started by establishing the design language for the collection: a consistent palette of teal/turquoise upholstery and powder-coated accents combined with polished chrome tubular and sheet metal frames. The collection was conceived to include a lounge chair, a long bench, a side table, a coffee table, a cylindrical side stool, and a floor lamp — all sharing the same visual identity. During this first week, I developed the initial 3D models for each piece, establishing the primary proportions, structural logic, and material assignments. Early renders were produced to validate the overall collection aesthetic before moving into design refinement.		



Supervisor **CALIBER™**

Industrial Supervisor's signature & stamp:

Date: 30/4/2026

Internship Logbook

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Student Name	Adam Mohamed Bannani
Student ID	TP073524
Company Name	Caliber

Week: 7	Start Date: 3/5/2026	End Date: 7/5/2026
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Freelance furniture collection design — Phase 2 (design refinement & finalisation). The objective was to refine the furniture collection designs with a distinctive arch motif, produce final renders for client presentation, and deliver the complete design package to the creative studio. In the second week of this project, I focused on refining the design language of the collection by introducing a recurring arch motif — inspired by Islamic and Moroccan architectural forms — as a unifying decorative and structural element across all pieces. The arched details were integrated into the bench legs, side table panels, coffee table frame, and chair structure, giving the collection a distinctive character that aligned with the studio's renovation-focused brand identity. Each iteration was progressively refined: from the initial simplified forms to a final version where the arch elements were fully resolved and balanced across the entire collection. Final high-quality renders were produced showing the complete set arranged in a staged environment. The full design package — including 3D models, renders, and visual references — was delivered to the creative studio by the end of the week.
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Industrial Supervisor's signature & stamp:

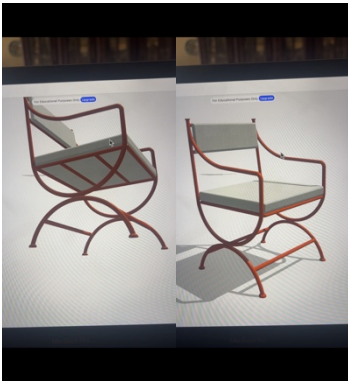


Date: 7/5/2026

Internship Logbook

Instruction:
Student is required to record the daily internship activities, learning and experience using this logbook. At the end of each week, the Industrial Supervisor needs to review the logbook and verify it. The completed logbooks must be submitted together with the internship reports.

Student Name	Adam Mohamed Bannani
Student ID	TP073524
Company Name	Caliber

Week: 8	Start Date: 10/5/2026	End Date: 14/5/2026
3D modelling and manufacturing of a custom curved iron chair with scrolled X-frame base. The objective was to design a chair to customised client measurements, model the complex curved tubular frame geometry, produce technical drawings, and oversee the fabrication of 4 units.		
This week, I was tasked with designing a chair built to a client's customised measurements. The design is a classically inspired armchair featuring a sculptural tubular frame finished in a terracotta/rust powder coat, with an upholstered seat and backrest in a light grey fabric. The most technically demanding aspect of this chair was the base: a curved, scrolled X-frame structure where the legs sweep outward in elegant curves and cross at the centre, connected by a horizontal stretcher. This required careful modelling of the tube bending paths in Shapr3D to ensure the curves were geometrically consistent and manufacturable. After finalising the 3D model with all custom dimensions, I produced the technical drawings and passed the files to the blacksmith team for tube bending and welding. I supervised the manufacturing process to ensure the complex frame geometry was executed accurately. A total of 4 chairs were produced for the order.		
		

Industrial Supervisor's signature & stamp:

Date: 14/5/2026

A handwritten signature in blue ink, followed by a blue rectangular stamp with the word "CALIBER" in bold, uppercase letters and a small trademark symbol (TM) to its right.

Internship Logbook

Instruction:
Student is required to record the daily internship activities, learning and experience using this logbook. At the end of each week, the Industrial Supervisor needs to review the logbook and verify it. The completed logbooks must be submitted together with the internship reports.

Student Name	Adam Mohamed Bannani
Student ID	TP073524
Company Name	Caliber

Week: 9	Start Date: 14/5/2026	End Date: 14/5/2026
3D modelling of new furniture pieces for upcoming production. The objective was to develop detailed 3D models for a lounge chair with an angular wooden X-frame base and a dining table with a tapered pyramid base, ready for technical drawing extraction and future manufacturing.		
The final week of the internship was dedicated to 3D modelling new furniture pieces in preparation for upcoming orders and production. The first piece was a lounge chair inspired by mid-century design — featuring a low-profile upholstered seat and backrest in a cream/off-white fabric, set on a distinctive angular X-frame base constructed from dark solid wood. The base geometry required careful modelling of the angled legs and their joinery to ensure structural accuracy. Multiple views and iterations were modelled to refine the proportions and achieve the correct seat height and recline angle. The second piece was a dining table with a bold tapered pyramid base in dark metal/sheet steel, supporting a large rectangular top with polished chamfered edges. The base was modelled as a hollow truncated pyramid — a structurally efficient and visually striking form that required precise sheet metal geometry to allow for laser cutting and welding. Both models were completed to a standard ready for technical drawing extraction and handoff to production.		



Industrial Supervisor's signature & stamp:

Date: 14/5/2026